

$$(3b) \lceil -x \rceil = -\lfloor x \rfloor$$

[Proof:] Let, $\lceil -x \rceil = p$. By Defn. 12, $p \in \mathbb{Z}$.

By (1b), $p-1 < -x < p \Rightarrow -1 \cdot (p-1) > -1 \cdot (-x) > -1 \cdot p$
 $\Rightarrow \boxed{\lceil p+1 \rceil > x > -p}$. There does not exist any integer in between $-p+1$ and $-p$ since $(-p+1) - (-p) = 1$ is the smallest absolute difference between any two arbitrary different integers. $\therefore -p$ is the largest integer smaller than or equal to x . By Defn. 12, $\lfloor x \rfloor = -p \Rightarrow p = -\lfloor x \rfloor \Rightarrow \boxed{\lceil -x \rceil = -\lfloor x \rfloor} \square$

$$(4a) \lfloor x+n \rfloor = \lfloor x \rfloor + n$$

[Proof:] By Defn. 12, if $\lfloor x+n \rfloor = y$, then $y \in \mathbb{Z}$.

$$\text{By 1(a), } y < x+n < y+1 \quad \begin{array}{l} x+n < y+1 \Rightarrow x < (y-n)+1 \\ y < x+n \Rightarrow (y-n) < x \end{array}$$

There does not exist any integer in between $(y-n)$ and $(y-n)+1$ since $(y-n)+1 - (y-n) = 1$ is the smallest absolute difference between two different arbitrary integers. $\therefore (y-n)$ is the largest integer smaller than or equal to x . By Defn. 12, $\lfloor x \rfloor = y-n$
 $\Rightarrow \boxed{\lfloor x \rfloor + n = y \Rightarrow \lfloor x+n \rfloor = \lfloor x \rfloor + n} \square$

$$(4b) \lceil x+n \rceil = \lceil x \rceil + n$$

[Proof:] Let, $\lceil x+n \rceil = y$. By Defn. 12, $y \in \mathbb{Z}$.

$$\text{By 1(b), } y-1 < x+n < y \quad y-1 < x+n \Rightarrow (y-n)-1 < x$$

There does not exist any integer in between $(y-n)$ and $(y-n)-1$ since $(y-n) - ((y-n)-1) = 1$ is the smallest absolute difference between two different arbitrary integers. $\therefore (y-n)$ is the smallest integer greater than or equal to x . By Defn. 12, $\lceil x \rceil = y-n \Rightarrow \lceil x \rceil + n = y$
 $\therefore \boxed{\lceil x+n \rceil = \lceil x \rceil + n} \square$

An useful approach for considering statements about the floor function is to let $x = n + \varepsilon$, where $n = \lfloor x \rfloor$ is an integer, and ε , the fractional part of x , satisfies the inequality $0 \leq \varepsilon < 1$.